Analytical Chemistry
Cumulative Exam
March 2, 2018

1. (20 pts) Consider a beaker containing 10 mL of 0.15 F acetic acid (pKa = 4.75).
   a) Calculate the pH of the solution
   b) Calculate the pH after adding to the beaker 5 mL of 0.1 M NaOH to the beaker.

2. (30 pts) Consider a beaker containing 10 mL of 0.15 F trisodium citrate,
   (pK_{a1} = 3.128,
   pK_{a2} = 4.76, pK_{a3} = 6.396)
   a) Abbreviate the salt as Na_3Ct, and write the K_{b1}, K_{b2}, and K_{b3} reactions.
   b) Calculate the numerical values for K_{b1}, K_{b2}, and K_{b3} and indicate the dominate reaction.
   c) Calculate the pH of the solution
   d) Calculate the pH after adding to the beaker 5 mL of 0.1 M HCl to the beaker.

3. (10 pts) Considering the answer to 1) and 2) above, which buffer has the greater initial buffer capacity?

4. (40 pts) Just consider Na_3Ct in water. For problem 2a, you wrote the reactions pertaining to trisodium citrate, use these to write:

   a) The equilibrium expressions for each reaction

   b) Charge Balance (don't forget sodium) and

   c) Mass Balance for sodium citrate

   d) Correctly, derive (DO NOT SOLVE) an equation that could be used to find the pH of this system. Your final equation may have $[OH^-]$ and $[Ct^{3-}]$ as variables. If you did it right, your equation will reduce to

$$[OH^-] = \sqrt{[Ct^{3-}]K_{b1}}$$

Equations

$$pH = pK_a + \log\frac{f_b}{f_a}$$ or $$Ka = \frac{[H^+]fb}{fa}$$

$$[OH^-] = \frac{-K_b \pm \sqrt{K_b^2 + 4K_b f_b}}{2} \approx \sqrt{FbK_b}$$

$$[H^+] = \frac{-K_a \pm \sqrt{K_a^2 + 4K_a f_a}}{2} \approx \sqrt{FaKa}$$

$$pH + pOH = 14$$
$$Kw = 10^{-14} = [H^+][OH^-]$$
$$KaKb = Kw$$